



IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 01 ASSIGNMENT

Cybersecurity Fundamentals & Security Concepts

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Assigned To	Cyber Security Analyst Interns
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Introduction

As part of Week 01 of the Cyber Security Analyst Internship Program at IT-Simplera Institute, this report covers the fundamentals of cybersecurity that every analyst needs to understand before moving into more advanced topics. Since my internship domain is Red Team / Offensive Security, I have tried to connect each task to how an offensive security professional would think about the same problem.

This report researches five real-world cyberattacks and analyzes them using the CIA Triad, reviews three cybersecurity tools that are directly relevant to red teaming, presents a secure home network design, and summarizes what a cybersecurity analyst typically does on a day-to-day basis. Completing this assignment helped me connect theoretical concepts from my coursework to practical, real-world scenarios.

Task 1 & 2: Real-World Cyberattacks and CIA Triad Analysis

The CIA Triad - Confidentiality, Integrity, and Availability - is the foundational model used to evaluate information security. **Confidentiality** ensures data is only accessible to authorized parties, **Integrity** ensures data remains accurate and unaltered, and **Availability** ensures systems and data are accessible when needed. Below are five real-world cyberattacks I researched, along with which part(s) of the CIA Triad each attack violated.

WannaCry Ransomware Attack (2017)

- **Organization(s) Affected:** NHS (UK), Telefonica, FedEx, and 200,000+ systems in 150 countries
- **Attack Type:** Ransomware / Worm
WannaCry was a self-propagating ransomware worm that exploited the EternalBlue vulnerability in Microsoft's SMB protocol (MS17-010). Once inside a network, it encrypted files on infected machines and demanded Bitcoin payment for decryption, spreading automatically to other vulnerable machines without any user interaction.
- **Impact:** Over 200,000 computers across 150 countries were infected within days. The UK's National Health Service (NHS) was hit hardest, forcing the cancellation of thousands of appointments and surgeries. Total global damages were estimated at around \$4 billion.
- **CIA Triad Violated: Availability**
By encrypting files and locking users out of their own systems, WannaCry directly violated Availability - legitimate users (including hospital staff) could not access critical systems and data when needed.

Equifax Data Breach (2017)

- **Organization(s) Affected:** Equifax (credit reporting agency, USA)

- **Attack Type:** Data Breach (Web Application Vulnerability Exploitation)
Attackers exploited an unpatched vulnerability in the Apache Struts web framework used by Equifax's online dispute portal. The flaw allowed remote code execution, giving attackers access to internal databases containing sensitive consumer records for over two months before detection.
- **Impact:** Personal data of approximately 147 million people was exposed, including Social Security numbers, birth dates, addresses, and driver's license numbers. Equifax paid over \$700 million in settlements and suffered severe reputational damage.
- **CIA Triad Violated: Confidentiality**
The breach exposed highly sensitive personal and financial data to unauthorized parties, which is a direct violation of Confidentiality - information meant to be accessible only to authorized individuals was stolen.

SolarWinds Supply Chain Attack (2020)

- **Organization(s) Affected:** SolarWinds and its customers, including US government agencies and Fortune 500 companies
- **Attack Type:** Supply Chain Attack / Advanced Persistent Threat (APT)
State-sponsored attackers compromised SolarWinds' software build system and inserted malicious code (the 'Sunburst' backdoor) into legitimate updates of the Orion IT management platform. When customers installed the trojanized update, attackers gained covert backdoor access to their networks.
- **Impact:** Around 18,000 organizations downloaded the compromised update, including several US federal agencies (Treasury, Commerce, Homeland Security). Attackers had months of undetected access, allowing extensive espionage and data theft.
- **CIA Triad Violated: Confidentiality and Integrity**
The attack violated Integrity because malicious code was secretly inserted into trusted software updates, and Confidentiality because attackers used the backdoor to spy on sensitive government and corporate data.

Colonial Pipeline Ransomware Attack (2021)

- **Organization(s) Affected:** Colonial Pipeline (largest fuel pipeline operator in the USA)
- **Attack Type:** Ransomware Attack
The DarkSide ransomware group gained access to Colonial Pipeline's network through a compromised VPN account password that lacked multi-factor authentication. The attackers encrypted business systems, prompting the company to proactively shut down pipeline operations as a precaution.

- **Impact:** The pipeline, which supplies nearly half of the US East Coast's fuel, was shut down for six days, causing fuel shortages, panic buying, and price spikes. Colonial Pipeline paid a ransom of approximately \$4.4 million (part of which was later recovered by the FBI).

- **CIA Triad Violated: Availability**

The shutdown of pipeline operations meant the fuel distribution service was unavailable to millions of customers, making this primarily an Availability violation, even though the attack originated as a confidentiality compromise (stolen credentials).

Yahoo Data Breaches (2013-2014 (disclosed in 2016-2017))

- **Organization(s) Affected:** Yahoo Inc.
- **Attack Type:** Data Breach (Credential/Account Compromise)

Yahoo suffered two separate massive breaches. State-sponsored hackers stole data using forged cookies and weak encryption (MD5-hashed passwords), gaining unauthorized access to user account databases over an extended period without detection.

- **Impact:** All 3 billion Yahoo user accounts were ultimately affected, making it one of the largest data breaches in history. Stolen data included names, email addresses, phone numbers, and security questions. The breach also reduced Yahoo's acquisition price by Verizon by \$350 million.

- **CIA Triad Violated: Confidentiality**

Unauthorized access to billions of user accounts and personal details is a clear Confidentiality violation, as private user information was exposed to attackers without consent.

Task 3: Cybersecurity Tools

Below are three cybersecurity tools I researched that are especially relevant to red team and offensive security work, which I also practice in my home lab using VirtualBox and Kali Linux.

Nmap (Network Mapper)

- **Purpose:** Network discovery and security auditing
- **Function:** Nmap scans networks to discover live hosts, open ports, running services, and operating systems. It uses techniques like TCP SYN scanning, service version detection, and OS fingerprinting to build a map of a target network.
- **Use Case (My Practice):** As a red team / penetration testing student, I use Nmap in the reconnaissance phase to identify live devices on my home lab network and check which ports/services are exposed before attempting further testing.

Wireshark

- **Purpose:** Network protocol analysis and packet capture

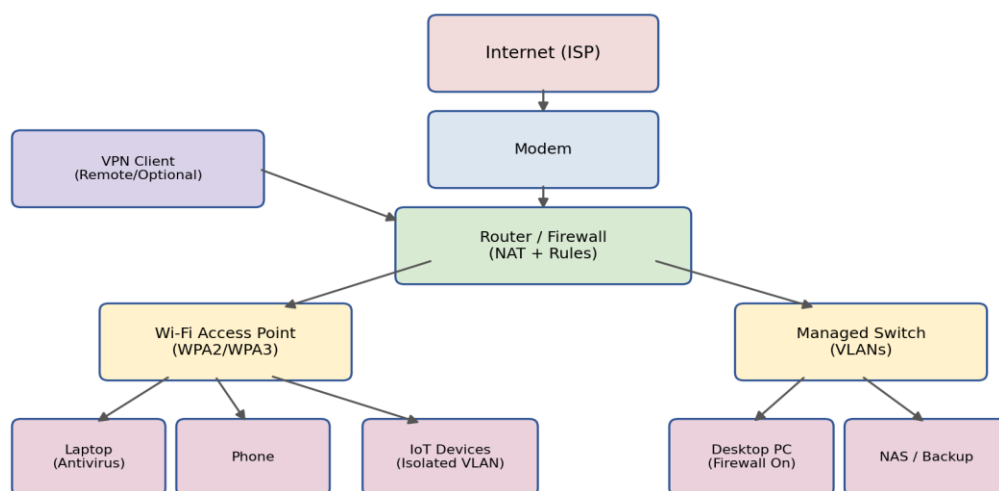
- **Function:** Wireshark captures live network traffic and lets analysts inspect individual packets in detail, following protocols like TCP, HTTP, and DNS. It can decrypt and dissect WPA2 handshakes when the correct keys are supplied, making it useful for wireless security analysis.
- **Use Case (My Practice):** I use Wireshark to analyze pre-captured WPA2 handshake files and general network traffic in my home lab, helping me understand how data flows across a network and how attacks/anomalies appear at the packet level.

Metasploit Framework

- **Purpose:** Penetration testing and exploit development
- **Function:** Metasploit provides a library of known exploits, payloads, and auxiliary modules that security professionals use to test whether a system is vulnerable to specific attacks. It also includes tools for post-exploitation and reporting.
- **Use Case (My Practice):** In a controlled lab environment (VirtualBox/Kali Linux), I use Metasploit to practice exploiting intentionally vulnerable machines, which builds the offensive security skills needed for red team engagements.

Task 4: Secure Home Network Design

A secure home network should limit exposure to the internet, segment devices by trust level, and layer multiple defenses so that a single failure doesn't compromise the whole network. The diagram below shows my proposed design, including the internet connection, modem, router/firewall, connected devices, antivirus protection, firewall rules, and an optional VPN for secure remote access.



Key design decisions: the router/firewall sits directly behind the modem and performs NAT along with inbound/outbound traffic filtering. Wi-Fi is secured with WPA3 (or WPA2 as a minimum) encryption and a strong passphrase. Personal computers run antivirus/endpoint protection, IoT and smart devices are isolated on a separate VLAN or guest network to reduce the blast radius if one device is compromised, and an optional VPN provides encrypted remote access to the home network when needed.

Task 5: What Does a Cybersecurity Analyst Do Every Day?

A cybersecurity analyst's daily routine revolves around monitoring, detecting, and responding to threats while continuously improving an organization's security posture. A typical day usually includes the following activities:

- Monitoring SIEM dashboards (e.g., Splunk, QRadar) for alerts and unusual activity across the network.
- Triaging and investigating security alerts to determine whether they represent real threats or false positives.
- Reviewing logs from firewalls, servers, and endpoints to identify signs of intrusion or policy violations.
- Performing vulnerability scans and tracking remediation of identified weaknesses with IT teams.
- Keeping up with threat intelligence feeds and news about newly disclosed vulnerabilities (CVEs) relevant to the organization.
- Assisting in incident response when a security event is confirmed, including containment and reporting.
- Documenting findings, writing reports, and communicating risks clearly to both technical and non-technical stakeholders.
- For red team-focused analysts specifically: conducting authorized penetration tests, simulating attacker behavior, and reporting exploitable weaknesses before real attackers can find them.

Overall, the role blends technical monitoring with continuous learning, since the threat landscape changes constantly and analysts must adapt their defenses accordingly.